PromptLite-Seg: Lightweight Zero-Shot Prompted Segmentation on PASCAL VOC 2012

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Abstract

PromptLite-Seg studies zero-shot prompted segmentation on PASCAL VOC 2012. Given a weak prompt consisting of a center point and a bounding box, the goal is to segment the target object without training on the benchmark. The project implements a simple center-color baseline, a stronger SAM-free robust superpixel method, and a GPU SAM ViT-B comparison. It further introduces prompt-robustness, prompt-uncertainty, and paired statistical reliability analyses. On a 30-sample VOC 2012 subset, clean SAM ViT-B reaches 0.8325 mean IoU with point+box prompts, while box-only prompts surprisingly reach 0.8565. The paired bootstrap confidence interval for this box-only gain is [0.0080, 0.0421]. Under moderate prompt noise, box perturbation drops SAM to 0.6462 IoU, but point perturbation remains at 0.8507. This shows that prompt quality is not a single scalar: the bounding box is the dominant information channel in this benchmark.

1 Introduction

Prompted segmentation has become a practical interface for general-purpose vision systems. Classic interactive systems such as GrabCut showed that a loose user box can provide enough foreground and background evidence for object extraction [2]. Later deep interactive methods studied stronger point-based guidance, including extreme-point prompts in DEXTR [3], iterative click correction with mask guidance [4], and click encoding with vision transformers in SimpleClick [5]. More recently, the Segment Anything Model (SAM) demonstrated a general promptable segmentation interface that accepts points and boxes and transfers zero-shot to new tasks [6]. This project therefore asks a complementary question: how far can a transparent, training-free prompted segmentation algorithm go on real benchmark images, and how sensitive is SAM to the structure of its prompts?

The task follows the course-design direction of zero-shot prompt segmentation. The benchmark is PASCAL VOC 2012, a standard object recognition and segmentation benchmark with semantic labels and evaluation protocols [1]. For each selected validation sample, the largest semantic object component is converted into a binary target mask. The prompt consists of a tight bounding box and an interior center point derived from the distance transform of the target mask.

2 Method

2.1 Prompt Generation

PASCAL VOC semantic masks provide foreground labels from 1 to 20 and background label 0. For each sample, PromptLite-Seg selects the largest connected foreground component as the target instance. The bounding box is the tight rectangle around this component. The point prompt is the pixel with maximum distance-transform value inside the component, which approximates a stable human center click.

2.2 Center-Color Baseline

The baseline estimates the target appearance from a small disk around the center point. Pixels inside the prompt box are segmented if their RGB distance to this median color prototype is below an adaptive Otsu threshold. Small components are removed, holes are filled, and the connected component nearest to the prompt point is retained.

2.3 Robust Superpixel Prompt Segmentation

The robust method first runs the center-color baseline to obtain a conservative foreground seed. It then oversegments the image into SLIC superpixels, which were designed to provide efficient boundary-aware image primitives [7]. Each superpixel is represented by mean Lab color and normalized spatial coordinates. Foreground prototypes are estimated from superpixels intersecting the center-point disk, while background prototypes are estimated from superpixels outside the prompt box and along the box border.

Each superpixel receives a foreground score based on its relative distance to foreground and background prototypes, combined with a weak spatial prior centered at the prompt point. The final mask is the union of the conservative color seed and superpixel consensus across slightly perturbed boxes, followed by morphology and point-connected component selection.

3 Experimental Setup

The experiment uses a compact subset of PASCAL VOC 2012 validation samples [1] from the Hugging Face mirror `nateraw/pascal-voc-2012`. The subset contains 30 samples and 12 object classes. The evaluation reports Intersection over Union (IoU) and Dice score:

$$\mathrm{IoU} = \frac{|P \cap G|}{|P \cup G|}, \qquad \mathrm{Dice} = \frac{2|P \cap G|}{|P| + |G|}.$$

The reproducible commands are:

```
.\scripts\run_all.ps1
python scripts/download_voc_subset.py --count 30
python scripts/run_experiment.py --max-samples 30
python scripts/analyze_results.py
.\.venv-sam\Scripts\python.exe scripts/run_robustness_experiment.py ^
  --max-samples 30 --trials 2 --include-sam --device cuda
.\.venv-sam\Scripts\python.exe scripts/run_prompt_uncertainty_experiment.py ^
  --max-samples 30 --trials 2 --ensemble-size 5 --device cuda
python scripts/analyze_statistics.py
```

4 Results

The robust method improves mean IoU by 0.0562 absolute points and mean Dice by 0.0523. At the sample level, it wins on 23 samples, ties on 5 samples, and loses on only 2 samples relative to

Table 1: Aggregate results on the 30-sample VOC 2012 subset.

Method	Mean IoU	Std IoU	Mean Dice	Std Dice
Center-color baseline	0.5468	0.2323	0.6754	0.2122
Robust superpixel	0.6031	0.2111	0.7277	0.1895
SAM ViT-B point+box	0.8325	0.1484	0.9002	0.1044

the center-color baseline.

The optional GPU SAM ViT-B comparison reaches 0.8325 mean IoU and 0.9002 mean Dice. This confirms that the prompts generated by the benchmark pipeline are informative enough for a modern foundation segmentation model, while the robust superpixel method remains useful as a transparent low-resource baseline.

4.1 Prompt Robustness

To evaluate prompt sensitivity, the robustness benchmark perturbs both the center point and the bounding box. Mild perturbations shift the point and box by roughly 5–6% of object size; moderate perturbations use roughly 10–12%. Each non-clean severity is evaluated with two deterministic trials per sample.

Table 2: Robustness under prompt perturbation.

Severity	Method	Mean IoU	Mean Dice
clean	robust superpixel	0.6031	0.7277
mild	robust superpixel	0.5579	0.6927
moderate	robust superpixel	0.4568	0.5967
clean	SAM ViT-B noisy prompt	0.8325	0.9002
mild	SAM ViT-B noisy prompt	0.7731	0.8537
moderate	SAM ViT-B noisy prompt	0.6756	0.7795

The robustness benchmark reveals the main limitation of prompt segmentation. SAM has much higher absolute accuracy, but it still loses about 0.157 IoU under moderate prompt noise. A naive lightweight prompt repair strategy does not improve SAM: it reduces moderate-noise SAM IoU from 0.6756 to 0.5745. SAM's own score-based selection also underperforms the original noisy prompt at 0.6118. This negative result suggests that prompt repair should preserve uncertainty instead of collapsing a noisy prompt into a single mask-derived point and box.

4.2 Prompt Uncertainty Research Experiment

The robustness result motivates a more precise research question: which prompt channel actually carries the useful information, and can multiple plausible prompts recover performance after noise? This is directly connected to the broader interactive-segmentation literature: box-driven methods such as GrabCut emphasize localization and background exclusion [2], while click-driven methods emphasize corrective user feedback [4, 5]. The prompt uncertainty experiment therefore evaluates three clean prompt modalities, decomposes point noise and box noise, and tests multi-prompt selection around a noisy observation. For efficiency and reproducibility, SAM image embeddings are computed once per sample and reused for all prompt queries.

The first surprising finding is that box-only prompting is stronger than point+box prompting on this subset. The point prompt is not useless, but it can over-constrain SAM when the box

Table 3: Prompt modality, noise decomposition, and uncertainty selection with SAM ViT-B.

Setting	Condition	Mean IoU	Mean Dice
clean modality	point-only	0.4430	0.5303
clean modality	box-only	0.8565	0.9150
clean modality	point+box	0.8325	0.9002
moderate noise	point noise	0.8507	0.9116
moderate noise	box noise	0.6462	0.7520
moderate noise	point+box noise	0.6385	0.7416
moderate ensemble	single noisy prompt	0.6385	0.7416
moderate ensemble	score selection	0.6796	0.7801
moderate ensemble	consistency medoid	0.6581	0.7595
moderate ensemble	oracle best-of-six	0.7401	0.8284

already localizes the object well. The second finding is sharper: point perturbation barely hurts SAM, while box perturbation is the main failure mode. The third finding is constructive but less statistically secure. Under moderate point+box noise, score-based multi-prompt selection improves mean IoU from 0.6385 to 0.6796, and the oracle best-of-six reaches 0.7401. This gap estimates the remaining headroom for a calibrated prompt-quality estimator.

4.3 Statistical Reliability

To avoid relying only on point estimates, the project adds paired statistical analysis over sample-level deltas. Each comparison uses a paired bootstrap 95% confidence interval and a paired sign-flip permutation test. For prompt-noise and ensemble experiments with two trials per sample, trial results are averaged within each sample before testing, so the effective unit remains the image rather than the perturbation trial.

Table 4: Paired IoU effects with 95% bootstrap confidence intervals.

Comparison	Delta IoU	95% CI	p
Robust superpixel vs. center-color	0.0562	[0.0301, 0.0891]	0.00002
SAM box-only vs. point+box	0.0240	[0.0080, 0.0421]	0.00922
Moderate box noise vs. point noise	-0.2044	[-0.2655, -0.1474]	0.00002
Score selection vs. single noisy prompt	0.0411	[-0.0044, 0.1029]	0.15418
Oracle best-of-six vs. single noisy prompt	0.1017	[0.0593, 0.1589]	0.00002

The statistical layer strengthens three conclusions and weakens one. The lightweight robust superpixel improvement is not just an aggregate artifact: its confidence interval is entirely positive. The box-only SAM result is also supported by a positive paired interval. The box-noise penalty is large and stable. However, score-based multi-prompt selection is only a promising trend in the current subset, because its interval crosses zero and the sign-flip test is not significant. The oracle result remains significant, so the recoverable headroom exists, but the current selection rule is not yet a reliable solution.

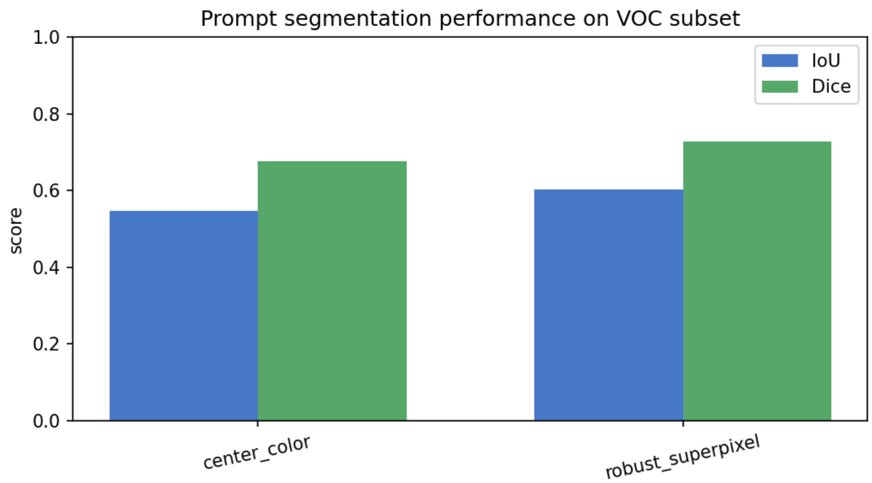

Figure 1: Aggregate IoU and Dice comparison.

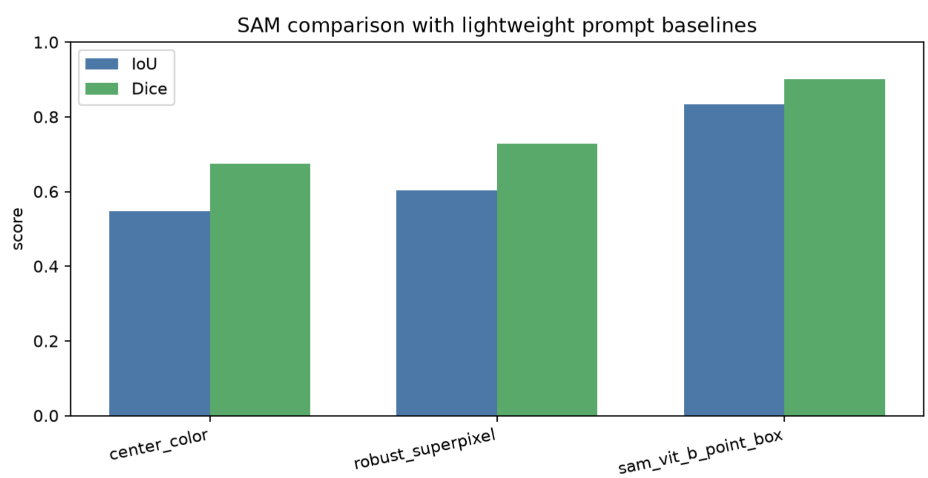

Figure 2: Comparison between lightweight prompt baselines and GPU SAM ViT-B.

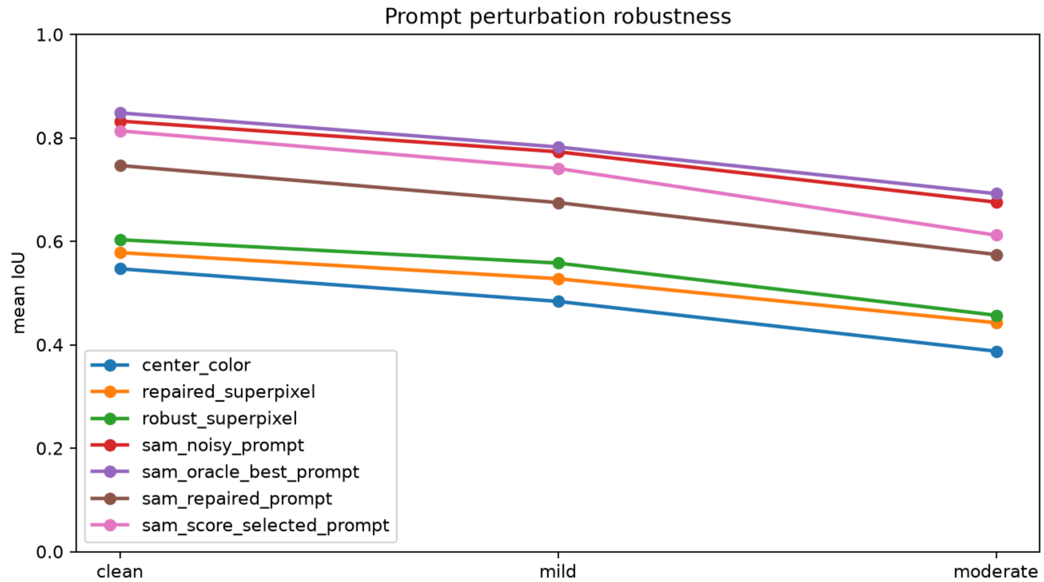

Figure 3: Prompt perturbation robustness under clean, mild, and moderate prompt noise.

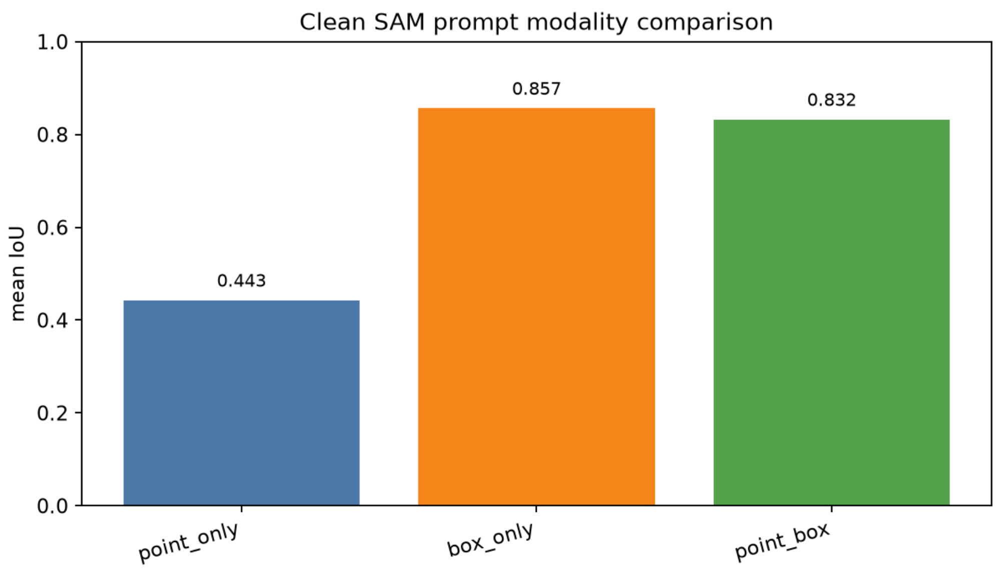

Figure 4: Clean SAM prompt modality comparison. Box-only prompting is strongest on the selected VOC subset.

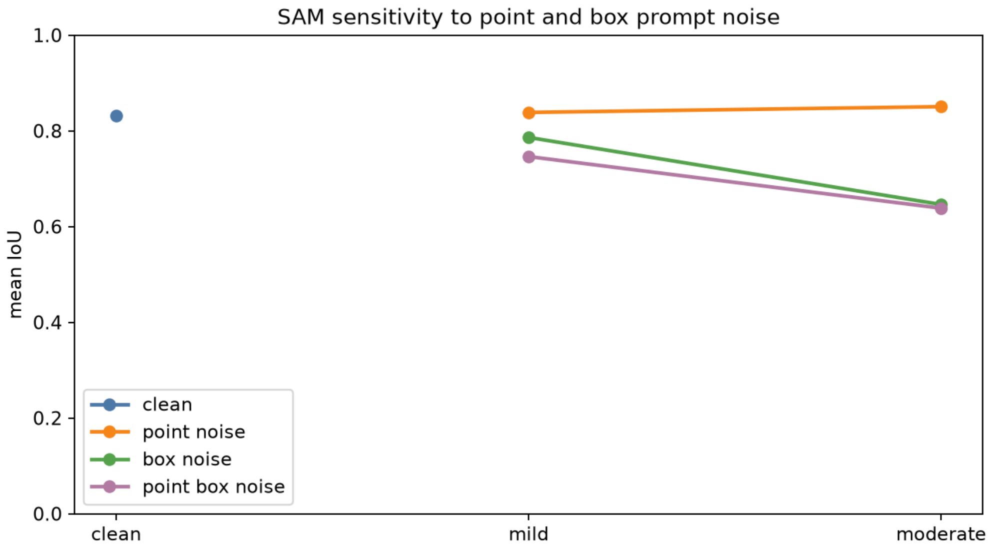

Figure 5: Noise decomposition. SAM is much more sensitive to box noise than point noise.

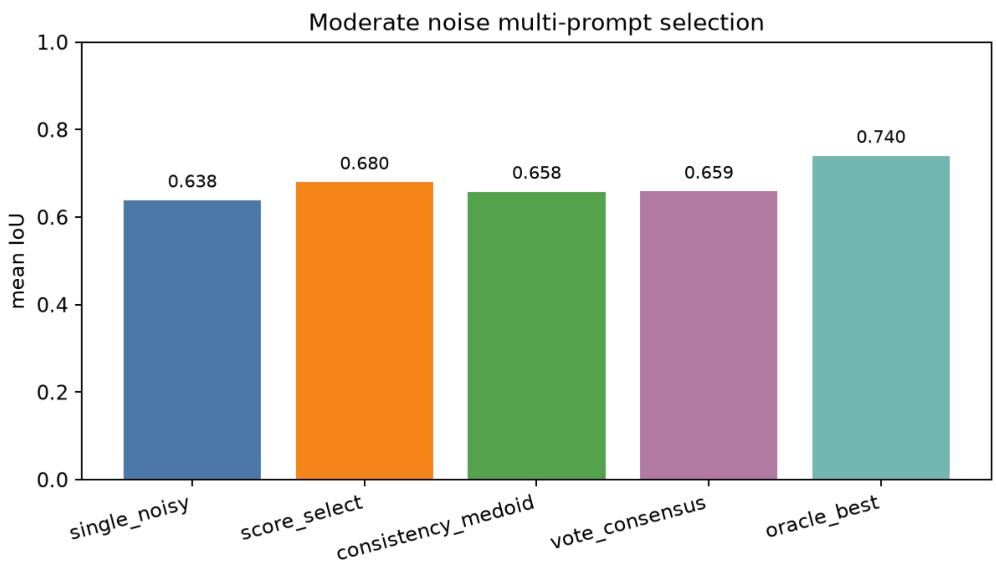

Figure 6: Multi-prompt selection under moderate point+box noise. Score selection recovers part of the lost IoU.

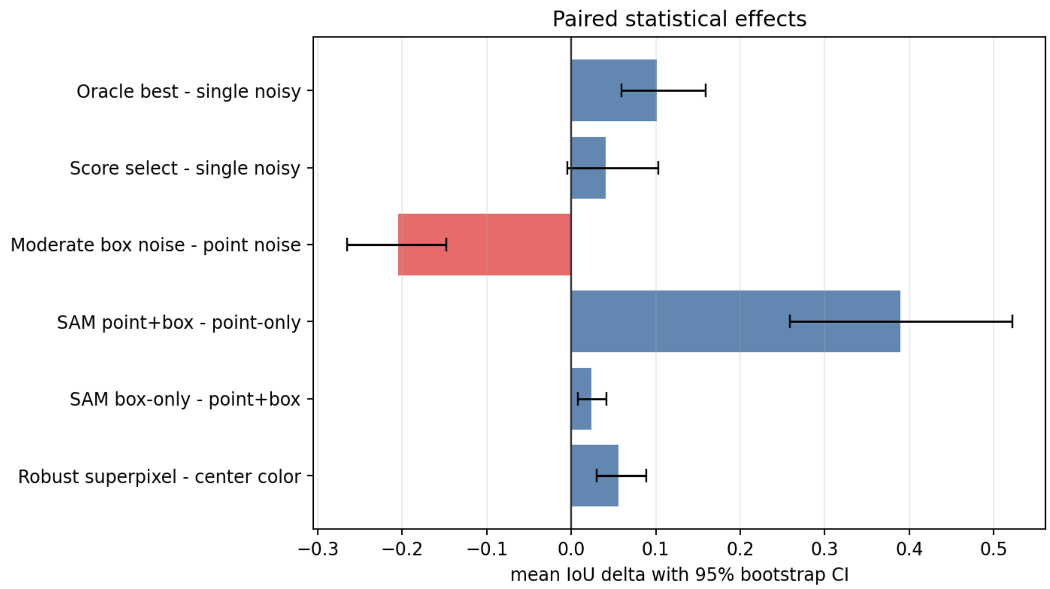

Figure 7: Paired IoU effects with 95% bootstrap confidence intervals.

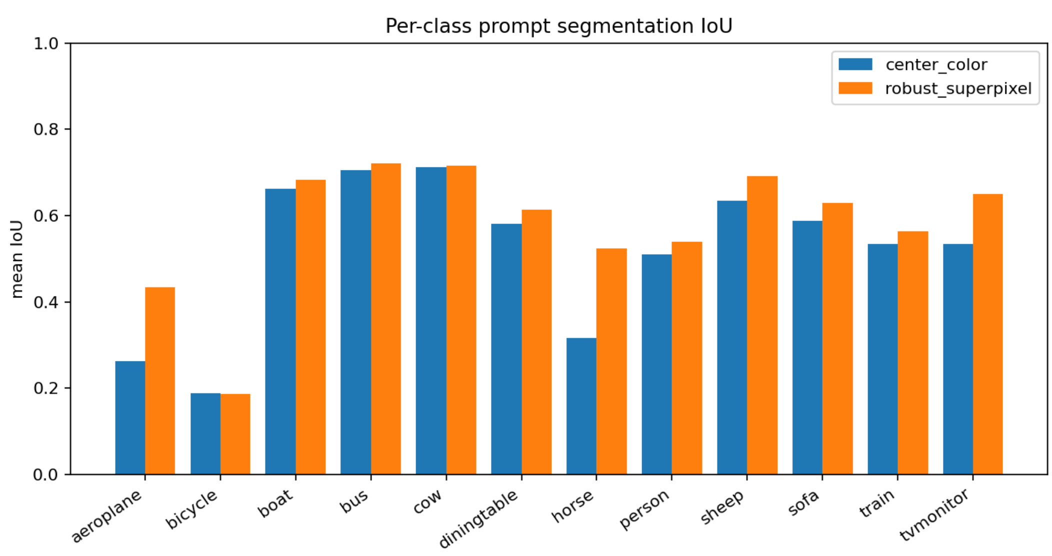

Figure 8: Per-class mean IoU. The strongest gains appear on horse, tvmonitor, and aeroplane examples.

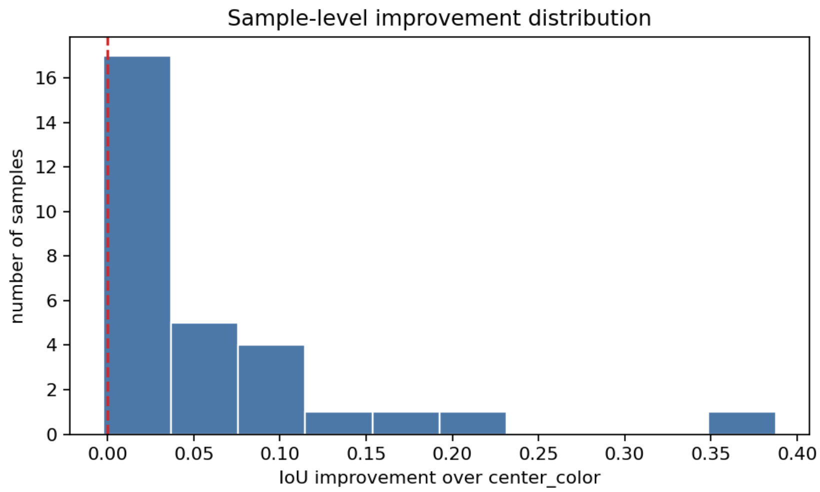

Figure 9: Distribution of sample-level IoU improvement over the center-color baseline.

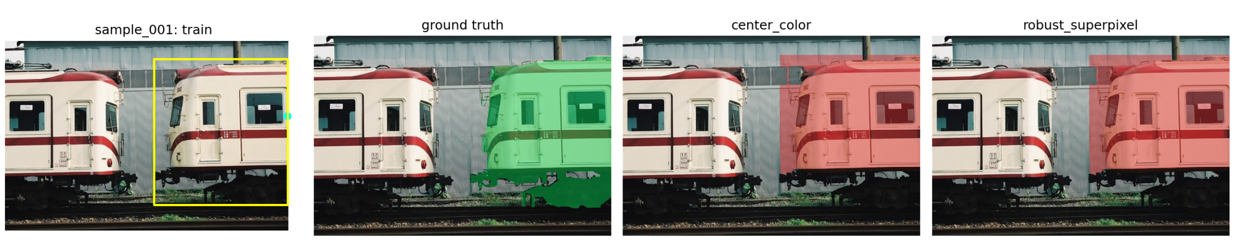

Figure 10: Qualitative example showing prompt, ground truth, center-color prediction, and robust superpixel prediction.

5 Analysis

The center-color baseline works when the target has a compact appearance and contrasts clearly with the local background. It fails when the object has multiple colors, when the center click lands on a non-representative part, or when background regions share similar colors.

The robust method improves the prompt interface in three ways. First, superpixels better respect local image boundaries than raw pixel thresholding. Second, background evidence from the box boundary and outside-box region suppresses leakage to nearby clutter. Third, box perturbation makes the prediction less sensitive to a single exact prompt.

The hardest cases include train, bicycle, diningtable, and boat images. Trains and dining tables often contain repetitive interior structures, bicycles contain thin parts that are easy to miss, and boats may include strong background colors inside the prompt box. The robustness experiment further shows that prompt noise is not a small implementation detail: it changes the ranking and reliability of methods.

The prompt uncertainty experiment deepens this conclusion. The original intuition might be that the point and box are simply two helpful prompts whose errors add smoothly. The measurements reject that assumption. In this benchmark, the box dominates localization, while the point behaves more like a weak disambiguation signal. The confidence intervals show that this is not merely a visual impression. Therefore, robust prompted segmentation should not treat all prompt noise equally. It should either protect the box channel, sample multiple plausible boxes, or learn a reliability model that predicts when SAM's internal score is well calibrated.

6 SAM Comparison and Lightweight Baseline

PromptLite-Seg is intentionally designed so that its main baseline remains SAM-free, CPU-friendly, deterministic, and easy to inspect. With GPU resources available, the project also evaluates SAM ViT-B using the same point-and-box prompts. This comparison is important because SAM is explicitly designed as a promptable foundation segmentation model [6], whereas PromptLite-Seg is closer in spirit to transparent classical interactive methods [2, 7]. SAM achieves much higher absolute accuracy, especially on objects with complex boundaries, while the lightweight method provides an interpretable baseline and a useful diagnostic fallback.

The deeper finding is that a stronger model is not automatically robust to prompt uncertainty. The oracle prompt selection significantly improves SAM under moderate noise, but the practical score-selection rule is not statistically confirmed on the current subset. This exposes a concrete future direction: robust prompted segmentation should model prompt uncertainty explicitly, for example by maintaining multiple plausible prompts or by learning a calibrated prompt-quality estimator.

7 Threats to Validity

The main limitation is dataset scale. The project uses a compact 30-sample subset of PASCAL VOC 2012 to keep the course artifact reproducible on a local machine. Paired bootstrap intervals reduce overinterpretation of sample-level differences, but they do not replace evaluation on the full validation set or on other datasets.

The prompt construction is also idealized. Points and boxes are derived from ground-truth masks rather than collected from human annotators. The perturbation benchmarks approximate annotation noise, but real user clicks and boxes may have different error distributions. Therefore, the box-dominance conclusion should be read as a finding for this controlled prompt-generation pipeline, not as a universal statement about all interactive segmentation settings.

The SAM experiments depend on a specific SAM ViT-B checkpoint and CUDA environment. Results may differ with SAM ViT-L, SAM ViT-H, SAM2-style models, or different preprocessing libraries. To make the artifact easier to reproduce, the repository includes a CPU-friendly default script, `scripts/run_all.ps1`, and keeps GPU experiments behind explicit flags.

Finally, the multi-prompt selection result is intentionally reported with caution. Score selection improves mean IoU, but its confidence interval crosses zero on the current subset. The significant oracle gain shows that recoverable headroom exists, but a reliable automatic selector remains future work rather than a completed claim.

8 Conclusion

This project implements and evaluates a zero-shot prompted segmentation pipeline on PASCAL VOC 2012. It satisfies the course requirement of running an algorithm on a benchmark, and it adds analysis beyond aggregate scores through per-class, success/failure, SAM comparison, prompt robustness, prompt uncertainty, and statistical reliability experiments. The main research conclusion is that prompt quality is a structured bottleneck: SAM is strong under clean prompts, box noise causes statistically stable degradation, point noise is comparatively harmless on this subset, and multi-prompt selection has measurable headroom but requires better calibration.

References

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